Beyond Current Initiatives:
ITMO Regulations to Ratchet up Ambition

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How a coalition of willing countries could make carbon trading serve the Paris Agreement instead of undermining it

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Where we are going

1 **The Paris Agreement**
What countries actually promised — and why it isn't enough

2 **What is an ITMO?**
Carbon trading between governments, and the trap hidden inside it

3 **A ladder of rules**
Five rungs of increasing ambition — the core proposal

4 **The alternatives**
Price floors, climate clubs, fossil fuel non-proliferation

1

The Paris Agreement

What 195 countries promised in 2015 — and the gap between the promise and the pledges

Why climate is the hardest cooperation problem we have

The benefit is global

A tonne of CO_2 avoided in Jakarta cools the planet exactly as much as a tonne avoided in Berlin. Everyone enjoys the benefit; only one country pays.

The cost is national

Closing a coal plant is paid by one treasury, one workforce, one electorate. Every government is tempted to let the others act first.

Consequence: there is no world government to impose a solution, so climate policy has to be built out of voluntary agreements between sovereign states.

Two things must be settled for any climate treaty to work:

(1) HOW MUCH total effort? **(2) WHO bears which share of it?**

Three decades of climate negotiations

1992
UNFCCC
Rio. Universal treaty. Splits the world in two: 'developed' pay, 'developing' receive.

1997
Kyoto Protocol
Binding targets, but only for rich countries. US never ratifies; Canada walks out.

2009
Copenhagen fails
The attempt to agree binding targets for all collapses. No treaty. Top-down is abandoned.

2015
Paris Agreement
Universal. A temperature target + voluntary national pledges (NDCs). No sanctions.

2024
COP29
Rulebook for Article 6 carbon trading finalised. New finance goal: $300 bn/yr by 2035.

Kyoto exempted developing countries and few signed. Copenhagen attempted a comprehensive agreement and collapsed. Paris let countries decide for themselves — and everybody signed.

What the Paris Agreement actually says

Two moving parts

1. A GLOBAL TARGET (Article 2)

"Holding the increase in the global average temperature to well below 2°C … and pursuing efforts to limit it to 1.5°C."

Agreed by everyone. Binding as an objective.

2. NATIONAL PLEDGES (NDCs)

Each country writes its own "Nationally Determined Contribution": how much it intends to cut, by when, in its own chosen units.

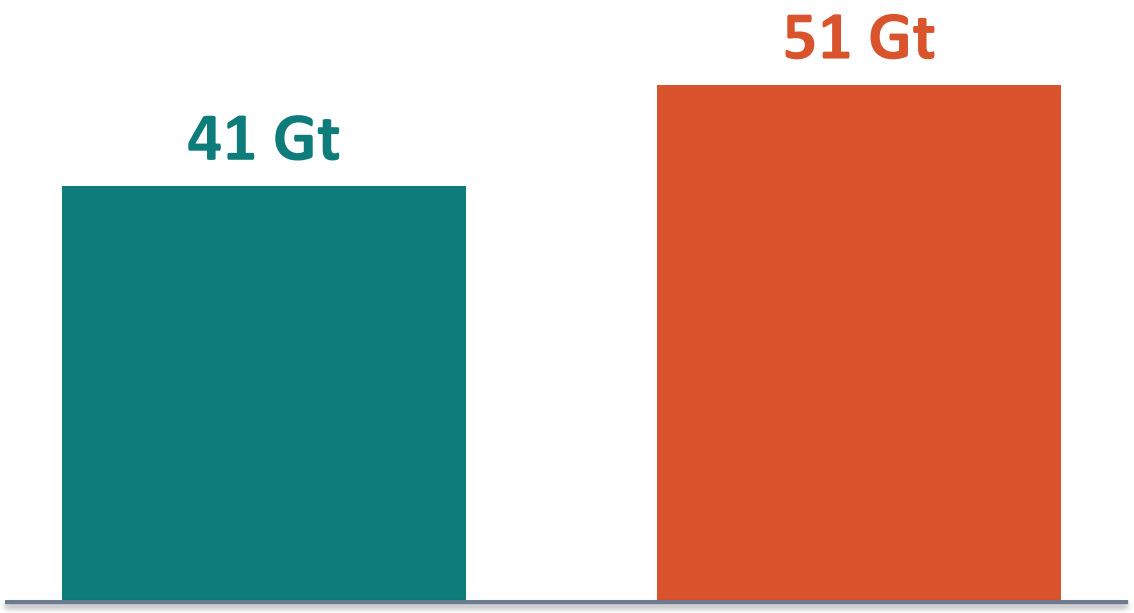
Submitting one is binding. Achieving it is not.

Nothing in the treaty connects the two.

- No formula tells a country what its fair share of the global target is.
- No penalty applies to a country that misses its own pledge.
- The only link is the "ratchet": every five years you submit a new pledge, not meant to be weaker.

It doesn't add up

The sum of the pledges is not the target



THE AMBITION GAP
≈ 10 Gt CO_2e / yr

Where that leads

+2.6°C

Projected warming by 2100 under current policies, according to Climate Action Tracker — and still rising after that.

This is not a failure of implementation. Even if every country perfectly delivered what it promised, we would still miss the target.

Why the gap exists: nobody could agree on who pays

The UNFCCC's only guidance on burden-sharing is a phrase from 1992:

**“Parties should protect the climate system (...)
on the basis of equity and in accordance with their
common but differentiated responsibilities and respective capabilities.” (CBDR)**

Everyone endorses it. Nobody agrees what it means. It embeds every possible principle:

Common action	Equal per capita	Historical responsibility	Ability to pay	Grandfathering
Every country takes on climate action	Every human has the same right to emit	A country owes what it emitted in the past	Effort in proportion to income	Cut in proportion to what you emit today

Because burden-sharing was never settled, no country can be told its pledge is too small.

Unresolved fairness → unenforceable ambition.

The other two pillars: money, and partnerships

Both useful. Neither caps emissions.

CLIMATE FINANCE

$300 bn / yr by 2035

Pledged at COP29, but loans count, and grant share is unspecified

26%

Of the current $100 bn goal, the share delivered as grants

$600 bn

What India asked for, in grants

$1.5 trillion in 2035

What an equal-per-capita carbon budget would imply

JUST ENERGY TRANSITION PARTNERSHIPS

A developing country commits to cut power-sector emissions; a group of rich countries lends on concessional terms.

Four signed so far

Indonesia · Vietnam · South Africa · Senegal ($2–20 bn each)

Three limits

Narrow sectoral scope
Loans, not grants, so no poverty reduction
Says nothing about China or HICs

Common weakness: finance and partnerships fund clean things. Neither puts a ceiling on dirty things.

2

What is an ITMO?

Carbon trading between governments — and the trap hidden inside it

First: why trading emissions is a good idea

The same tonne costs very different amounts to avoid

Country A

$200

to avoid one more tonne of CO_2



A pays B

Country B

$20

to avoid one more tonne of CO_2

If A pays B, say $60, to cut the tonne instead:

- the atmosphere sees the same tonne avoided
- A saves $140
- B earns $40 above its cost

Both are better off. Nothing was lost.

This is the entire economic case for carbon markets, and it is a good one.

But notice the hidden assumption: it works only if the TOTAL amount of cutting is already fixed and correct.

Trade reallocates effort efficiently. It cannot create effort that nobody committed to.

Article 6.2: how a transfer actually works

ITMO = Internationally Transferred Mitigation Outcome



The “corresponding adjustment”: the tonnes are subtracted from the buyer's emissions account and added to the seller's. That prevents double counting — the same tonne cannot be claimed twice.

- Rulebook finalised at **COP29 in 2024**. Purely bilateral: any two governments, any terms.
- **The accounting is watertight. Whether the underlying pledges were ambitious is not regulated at all.**

This is happening now, and it is not marginal

SWITZERLAND

First deals ever finalised, with Thailand and Ghana. May cover over 50% of its 2035 target with ITMOs.

EUROPEAN UNION

2040 law allows credits worth 5% of 1990 emissions — letting the EU emit domestically 50% above its 2040 goal.

JAPAN

Plans to emit up to 53% above its 2040 goal by buying abroad.

ALSO BUYING

South Korea, Norway, Singapore.

projected ITMO market by 2040:
500 $MtCO_2$ per year

That is roughly the annual emissions of Germany, moved on paper from one national account to another.

Meanwhile on the selling side: China pledges to cut 7–10% from its peak by 2035. Alignment with 2°C would need about 20%.

The trap, in numbers

A toy world with two regions

The planet can sustain 1,000 Gt in total. Without any climate policy, each region would emit 750 Gt.

	Region Rich	Region Poor	World total
Its own pledge (NDC)	500 Gt	1,000 Gt	1,500 Gt
Emissions WITHOUT trading	500	750	1,250 Gt

Poor has 250 Gt of spare room it never had to earn. It sells that room to Rich, very cheaply.

Emissions WITH trading	750	750	1,500 Gt

Trading made global emissions rise from 1,250 to 1,500 Gt.

The deep reason: the price is wrong

The sum of NDCs is too loose

There is an ambition gap of ~10 Gt/yr

→

Spare tonnes are abundant

Sellers hold headroom they never meant to use (“hot air”)

→

ITMOs are cheap

The price reflects the loose pledges, not the Paris target

→

Buying beats doing

Domestic transition is postponed in favor of cheap paper

A carbon price is supposed to be the shadow price of a binding constraint.

If the constraint isn't binding — if the pledges collectively allow more emissions than the target — then the market price measures nothing real. It is the price of a promise nobody had to keep.

Unregulated ITMOs do not export ambition from rich countries. They import laxity into them.

3

A ladder of rules

Five rungs of increasing ambition — for a coalition of the willing

What has already been proposed

Two influential fixes — and what they share

Michaelowa et al. (*Climate Policy*, 2019)

"ITMOs must be additional"

A supervisory board checks whether the seller's pledge is below business-as-usual. If yes, trade is allowed. If not, the pledge generates hot air, and each individual project must be audited for additionality.

La Hoz Theuer et al. (*Climate Policy*, 2019)

"Just cap the quantity of ITMOs"

They test four ways of defining business-as-usual and show that every one of them still creates hot air somewhere. So instead: simply limit sales of ITMOs to a fixed share of a country's emissions — 1% or 5%.

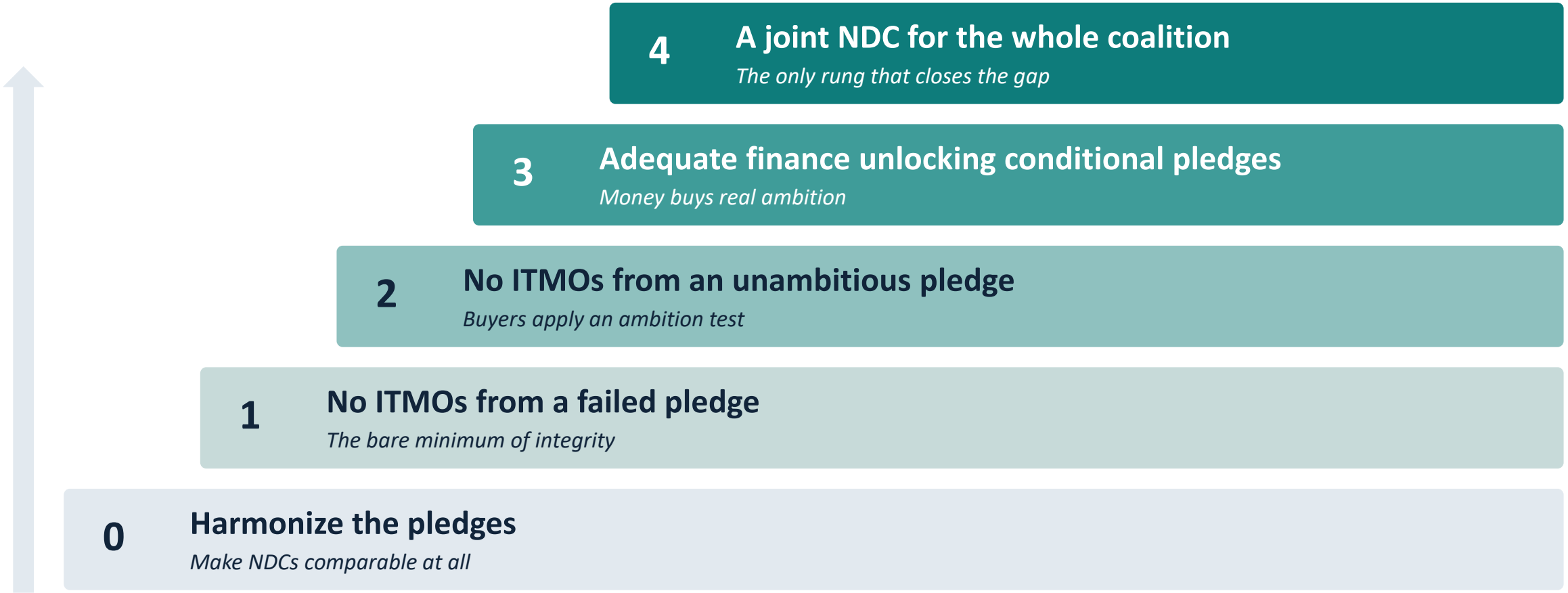
Both share one assumption:

That a pledge can be judged COUNTRY BY COUNTRY, against that country's own projected emissions.

I will argue that assumption is both unfair and ineffective — but first, the rules worth having anyway.

A ladder of five rules

Each rung is voluntary, and more demanding than the one below



Rungs 0–1 are almost costless. Rungs 2–3 a principled buyer can adopt alone. Rung 4 needs a coalition.

Rung 0 — Harmonize the pledges

Today, every country writes its pledge in its own way You cannot add them up without making assumptions.

Which greenhouse gases?	each country picks
Which sectors? Forests included?	each country picks
Absolute cut, or cut per unit of GDP?	each country picks
Which conversion factors between gases?	each country picks

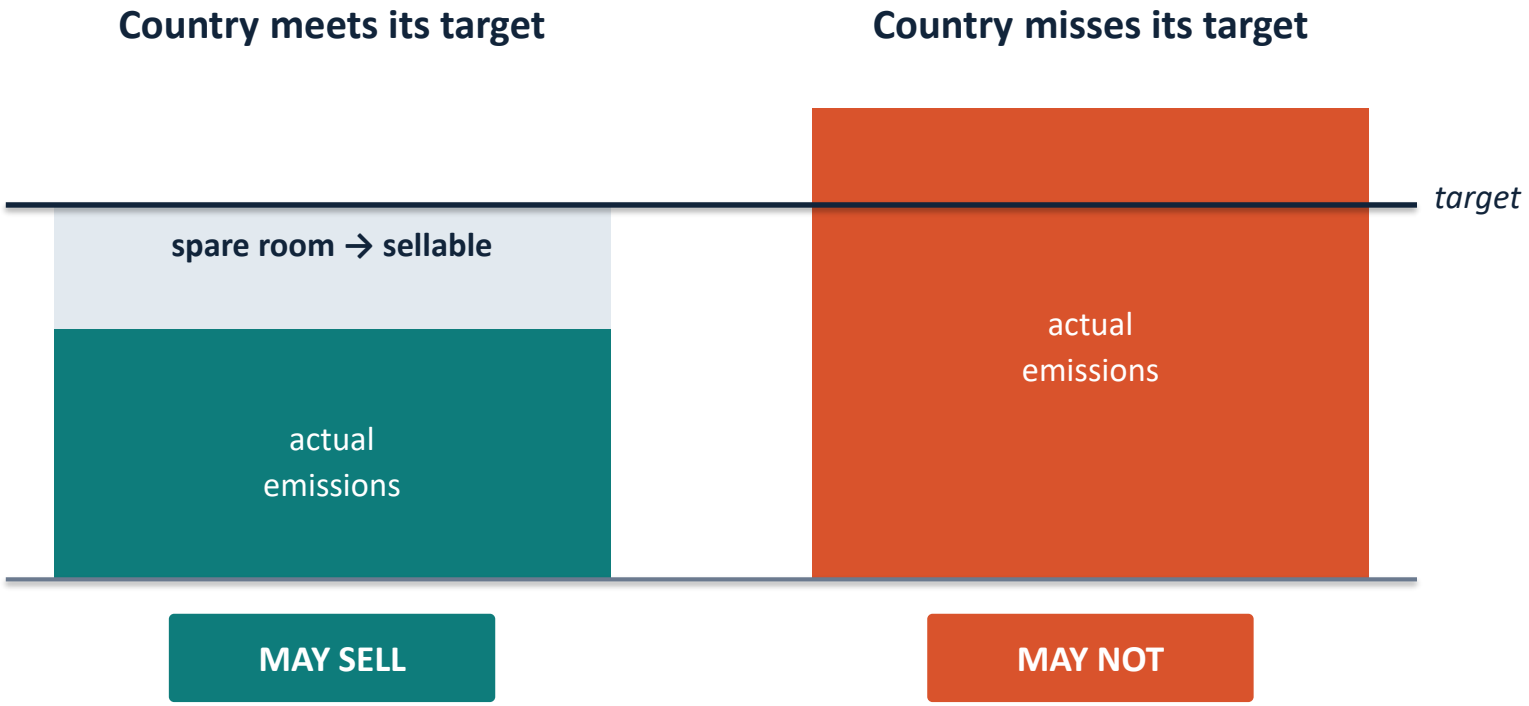
What we shouldrequire

- Cover all Kyoto gases, all sectors
- State an absolute target, in CO_2-equivalent
- One conversion factor: IPCC AR6 GWP100 or GWP500
- Separate land use from the rest
- Ideally: a cumulative carbon budget, a yearly path, and the intended use of ITMOs

Not glamorous, but a precondition: you cannot enforce a rule about ambition until ambition is measurable.

Rung 1 — No ITMOs from a failed pledge

A country may sell only the tonnes by which its emissions fall BELOW its own target.



Obvious once stated — yet nothing punishes countries from doing so.

It should also cover countries that quietly weakened an earlier target.

E.g. Angola doubled its assumed business-as-usual in a revised pledge, creating headroom out of thin air.

Rung 2 — Refuse to buy from an unambitious pledge

Meeting your pledge is not enough if the pledge itself was undemanding.
A principled buyer should only from a country that passes two tests:

Test A

Is the target below business-as-usual?

If a pledge requires no effort at all, it is not a pledge.

Test B

Is it below *either* cost-optimal 2°C emissions *or* an equal-per-capita share?

If it fails both, no burden-sharing principle can justify it.

Simple proxy

Are the country's emissions per person above the world average?

Crude, but avoids arguing about which 2°C model to use.

The first ITMO deal ever finalised — Switzerland buying from Thailand — passes all of these tests.

Yet the price paid is still too low, because the world's pledges still don't add up.

Rung 3 — Adequate finance unlocking conditional pledges

Most developing countries file two targets. The stronger one should apply only if promised finance arrives.

Unconditional target

"What we will do with our own resources." Often above business-as-usual — which under rung 2 would exclude the country from selling.

Conditional target

"What we will do if we receive $X of climate finance." Typically well below business-as-usual.

The rule: Assess a seller against its conditional target only if the promised finance has been delivered.

- **This turns market access into leverage.** To buy, you must first provide real, grant-based climate finance.
- Only "unidirectional" finance counts — not ITMO revenue itself, or the buyer would pay once and claim twice.
- Sellers could organise: a bloc of low emitters has real market power here.

Issues with national assessments of pledges

Two reasons why judging pledges one country at a time fails

1. Ineffective

Even if every seller's target is individually deemed ambitious enough, the pledges can still add up to more than the planet can sustain.

The ambition gap persists. Demand for ITMOs stays weak, the price stays low, and buyers keep postponing domestic transition.

2. Unfair

A low emitter may quite legitimately set its target *above* its own business-as-usual — and still deserve to sell. It would simply be claiming a fair share of the atmosphere that it is not yet using.

The Global South excluding China already asks for less than their fair share: 13.5 $GtCO_2$ in 2030, or 2.8t per person.

The ambition gap is meaningful at the scale of the world (or a large coalition).

That is, at the scale where there is consensus that climate action is inadequate, not at national scale.

Rung 4 — A coalition with a joint NDC

A group of willing countries submits one joint pledge, aligned with the Paris target, and commits to trade ITMOs only among themselves.

Why a JOINT pledge?

Ambition can finally be judged at the scale where it means something: does the coalition's total fit inside a Paris-aligned carbon budget? Yes or no.

Why a CLOSED market?

If members traded with outsiders holding loose pledges, weak ambition would flow straight back in. The boundary is what protects the price.

What it makes possible

Inside the coalition, ambition is guaranteed by construction — so members can trade freely, with no need for restrictions of rungs 2 and 3.

The precedent already exists: the European Union submits a single joint NDC for 27 member states.

How it would work in practice

Four design decisions

1	**The coalition's budget**	Proposed benchmark: an equal-per-capita share of a global budget giving a 75% chance of staying below 2°C. For a coalition of China, the Global South, Europe and Japan that is about 750 $GtCO_2$ over 2026–2100.
2	**A gentle start, a steepening path**	The 2026 target simply equals current emissions. 2027 sits slightly below business-as-usual. The wedge widens every year until the coalition reaches a Paris-aligned reduction rate.
3	**Sharing it out**	Start from a focal point: **each country gets the minimum of its projected emissions, its existing pledge, and its equal-per-capita share of the coalition's path**. Then hand extra room to the poorest members and to those hit hardest.
4	**A scientific council**	Projects baselines, models the distributional effects, proposes allocations. Every member may nominate a team; in case of disagreement, votes are weighted by the population represented.

Ideally the joint pledge sets a cumulative budget split into yearly national targets — exactly what a real cap-and-trade needs.

Is this remotely feasible?

Four reasons to think so

Coalitions beat consensus

The UNFCCC decides by consensus, so it moves at the speed of its most reluctant member. A parallel coalition does not.

The public is ahead of governments

Surveys across many countries find majority support for international climate policies — even when respondents are told what it would cost them personally.

It needs no new treaty

Everything here operates inside the Paris Agreement, using Article 6.2. It operationalizes CBDR rather than replacing it.

The door is already opening

COP30 launched the Open Coalition on Compliance Carbon Markets (China, the EU, Brazil and 15 others) and the Article 6 Ambition Alliance (10 countries, incl. Switzerland and Germany).

And if new members join? They may propose a new allocation, adopted by majority vote — provided it does not shrink emissions coverage or lower the projected ITMO price.

4
The alternatives

If not this, then what? Five competing proposals

At the EU level: leverage, not offset

Zachmann & Weisser (Bruegel Policy Brief, 2026)

The three moves

National credits

Issued at country level, not project by project.

Private buyers

Firms buy for their own compliance with domestic policies such as the EU ETS.

Leverage, not offset

A 1:5 multiplier: each tonne offset at home buys five abroad.

What it takes to implement

1 A mitigation value

What a credit is really worth: the seller's overachievement of its **baseline**, divided by the credits it sold.

Ex: Country sells 5 ITMOs but emits only 3 units below baseline → value = 3/5 = 0.6

2 A Carbon Credits Clearing House

Firms collateralise foreign credits to borrow domestic allowances, so they can buy ITMOs before the 2040 compliance date.

At 1:5 leverage and value 0.6 → 5 / 0.6 = 8.33 ITMOs needed per EU allowance

3 A Carbon Buyers' Club

Participating countries coordinate compliance standards and fix each seller's baseline.

e.g. baseline set by equal per capita, or by historical responsibility

Would allow 27 Gt of net mitigation

≈ a fifth of the global emissions gap

Right diagnosis: Article 6 as a lever for extra mitigation, not cost containment.
And their Buyers' Club is a close cousin of the coalition proposed here.

Differentiated carbon price floors

Parry, Black, Roaf (IMF, 2021); Wolfram et al. (2025)

Every country prices industrial carbon nationally and keeps the revenue. Minimum prices vary by income level:

$75 / t	**$50 / t**	**$25 / t**
high-income	upper-middle	lower-income

Wolfram et al. narrow this to steel, cement and aluminium, with a coalition-wide border tax at $75.

The problems

- It barely moves the needle: **global CO_2 falls by about 2%**.
- It asks the EU to drop its border adjustment in exchange for China pricing manufacturing — yet China is ~70% of covered emissions and will not accept $50/t.
- Almost no North–South transfer, and no gains from trade, because prices deliberately differ.

Deeper point: distributive effects of differentiated prices can be replicated by a uniform price with a suitable allocation of emission rights. Same winners and losers, without the efficiency loss — so rejecting monetary transfers is a costlier route to the same place.

Uniform price with transfers, or climate clubs: implausible ideals

Global cap-and-trade with revenue sharing

One price, one cap, and the revenue shared internationally on a roughly equal-per-capita basis. Proposed in many forms since 1990.

Strengths

Guarantees the total. Prices firms directly rather than trusting governments to legislate.

Obstacle

Governments fear the loss of sovereignty.

Climate clubs and refunding clubs

Reward countries that price carbon above a benchmark or emit below average; punish those that don't, with tariffs or forfeited refunds.

Problems

Rewards can flow to wealthy low-emitters such as Norway.
Enforcement depends on punishing countries agreeing to be penalised.
Refunding clubs need an up-front fund estimated at $2.6 trillion — 0.3% of world GDP over a decade.

The proposal in this talk aims at the same efficiency and fairness as a redistributive cap-and-trade, while leaving each country free to choose its own policies.

Fossil fuel non-proliferation

The most popular alternative — and, I argue, the weakest

A campaign backed by a million individuals, four thousand organisations and 101 Nobel laureates, calling for a binding plan to end coal, oil and gas expansion. The associated "Fair Shares Phase Out" sets country-specific end dates for oil extraction:

United States	Russia	Saudi Arabia	Iraq
2031	**2037**	**2041**	**2050**

It needs exactly the wrong people	It requires fossil fuel exporters to act — the countries least likely to.
It enriches resource owners	Cutting supply rather than demand raises fossil rents and generates no revenue to compensate low-income consumers for higher prices.
It puts extraction in the wrong places	Cheap oil from Qatar would be phased out 13 years before dirtier oil from Venezuela.

A producer-side carbon tax fixes the latter (efficiency) problem — but not the other two.

Everything on one page

Simplified from the paper's full comparison

International policy	Cuts emissions	Least cost	Fair to low-income	Politically acceptable
Unregulated ITMOs (status quo)	0	+	0	+++
ITMOs + country-level integrity	+	+	+	++
ITMOs avoiding the ambition gap	+++	+++	++	+
JETPs (status quo)	+	0	+	+++
JETPs with grants and wider scope	++	+	++	++
Differentiated price floors	+	+	–	++
Climate club / refunding club	+++	+++	–	++
Global cap-and-trade with transfers	++++	+++	+++	+
Fossil fuel non-proliferation	+	–	–	–
Expansion of climate finance	++	+	+	+++

+++ strong, 0 neutral, – negative. The paper's full table scores nine criteria across nineteen policies, including acceptability to oil states.

5
Takeaways

What follows from all this

Five things to take away

1

The problem is not cheating, it is arithmetic

Even under perfect compliance, the pledges add up to 2.6°C. That is a design failure, not an enforcement failure.

2

Carbon trading is a good idea attached to a broken constraint

Trade allocates a fixed effort efficiently. When the total effort is wrong, trade spreads the laxity instead of the ambition.

3

Some rules cost almost nothing and should be adopted now

Harmonize pledges; never buy from a country that is missing its own target. Neither requires a coalition or a new treaty.

4

Only a joint pledge fixes the issue

Judging ambition country by country is unfair and ineffective. Move the unit of assessment to a coalition.

5

The alternatives are worse, with one exception

Price floors, clubs and non-proliferation lack efficiency, fairness or realism. Expanded JETPs are the exception.

Carbon trading will happen either way.
The only question is whether a credible ceiling binds it.

A tightly regulated ITMO coalition, plus ambitiously expanded Just Energy Transition Partnerships, is a viable pathway to international decarbonization.

Thank you

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sciencedirect.com/science/article/pii/S0959378026001068

More complete preprint: github.com/bixiou/global_tax_attitudes → paper/itmo_rules.pdf